Two exact counterexamples to backtracking PageRank monotonicity

MathDB 356988

Principia Math In Progress

September 18, 2026

Abstract

On the house graph with uniform personalization, one vertex's μ-PageRank decreases and then increases. A six-vertex theta graph has PageRank spread strictly increasing for $0 < \mu < 1$ and strictly decreasing for $\mu > 1$. These give exact counterexamples to both monotonicity assertions in Conjecture 3.1 of the cited preprint, and also to the opposite spread direction suggested by the source's prose. All graphs are connected, simple, undirected, and have minimum degree two. Rational stationary distributions provide certificates requiring only polynomial arithmetic.

Task 1. Complete counterexamples

We use Definition 2.4 of [1] exactly. For a graph with n vertices, replace each edge by its two orientations. Write y_{ij} for the stationary mass of oriented edge (i, j), and d_i for the degree of vertex i. Fix damping $0 < \epsilon < 1$, backtracking weight $\mu \geq 0$, and uniform vertex personalization. Then

$$u_{ij} = \frac{1}{nd_i}, \qquad y_{ij} = \frac{1-\epsilon}{nd_i} + \frac{\epsilon}{d_i - 1 + \mu}\left(\mu y_{ji} + \sum_{h\sim i,\ h\neq j} y_{hi}\right). \tag{1}$$

The vertex PageRank is the source's tail projection

$$\pi_\mu(i) = \sum_{j\sim i} y_{ij}. \tag{2}$$

Every graph below has $d_i \geq 2$, so (1) is defined even at $\mu = 0$. If P_μ is the column-stochastic transition matrix on oriented edges, (1) is $y = \epsilon P_\mu y + (1-\epsilon)u$. It has the unique solution

$$y = (1-\epsilon)\sum_{r=0}^{\infty} \epsilon^r P_\mu^r u, \tag{3}$$

because the series converges in the ℓ^1 norm. This solution is positive and has total mass one. Thus a normalized rational vector satisfying (1) certifies the exact PageRank.

Theorem 1. *The assertion that every coordinate of π_μ is monotonic in μ is false, even for the five-vertex house graph with uniform personalization and $\epsilon = 1/2$.*

Proof. Take vertices $0, 1, 2, 3, 4$ and edges

$$E_H = \{01, 03, 04, 12, 23, 34\}.$$

Vertices $0, 3$ have degree three, and $1, 2, 4$ have degree two. The reflection interchanges $0 \leftrightarrow 3$ and $1 \leftrightarrow 2$, fixing 4. Let

$$D = 9\mu^3 + 87\mu^2 + 263\mu + 241 > 0.$$

The six directed-edge orbits have the following masses; each row lists the numerator of the common denominator $30D$.

Mass	Directed edges	Numerator of the mass
a	$03, 30$	$6(\mu+5)(3\mu^2+16\mu+19)$
b	$01, 32$	$3(7\mu^3+67\mu^2+201\mu+181)$
c	$04, 34$	$21\mu^3+199\mu^2+595\mu+553$
d	$10, 23$	$6(4\mu^3+39\mu^2+120\mu+113)$
f	$12, 21$	$3(9\mu^3+85\mu^2+247\mu+211)$
g	$40, 43$	$2(12\mu^3+115\mu^2+346\mu+319)$

For a direct verification of the table, substitute its entries in the six instances of (1):

$$a = \frac{1}{30} + \frac{\mu a + d + g}{2(\mu+2)}, \qquad b = \frac{1}{30} + \frac{a + \mu d + g}{2(\mu+2)},$$
$$c = \frac{1}{30} + \frac{a + d + \mu g}{2(\mu+2)}, \qquad d = \frac{1}{20} + \frac{\mu b + f}{2(\mu+1)},$$
$$f = \frac{1}{20} + \frac{b + \mu f}{2(\mu+1)}, \qquad g = \frac{1}{20} + \frac{c}{2}.$$

All six identities follow by multiplication by their denominators; also $2(a+b+c+d+f+g) = 1$. Positivity of the displayed coefficients proves nonnegativity of every mass. Uniqueness in (3) identifies this vector with the stationary distribution.

The roof vertex 4 consequently has

$$p(\mu) := \pi_\mu(4) = 2g = \frac{2(12\mu^3+115\mu^2+346\mu+319)}{15D}. \tag{4}$$

In particular,

$$p(1/2) = \frac{8356}{47445}, \qquad p(1) = \frac{22}{125}, \qquad p(2) = \frac{3134}{17805},$$

and the exact positive differences are

$$p(1/2) - p(1) = \frac{142}{1186125}, \qquad p(2) - p(1) = \frac{8}{445125}. \tag{5}$$

Thus the coordinate is neither nondecreasing nor nonincreasing, using three strictly positive backtracking parameters. □

For additional information, differentiation of (4) gives

$$p'(\mu) = \frac{2(9\mu^4+84\mu^3+206\mu^2-76\mu-511)}{15D^2}.$$

Its numerator has exactly one positive zero: there is one sign change in its coefficient sequence, while it is negative at 0 and eventually positive. Hence this coordinate has exactly one minimum for $\mu > 0$.

Theorem 2. *The PageRank spread need not be monotonic in either direction. With uniform personalization and $\epsilon = 9/10$, the graph*

$$E_\Theta = \{01, 04, 05, 12, 23, 34, 35\}$$

on $\{0, 1, 2, 3, 4, 5\}$ has spread strictly increasing on $[0, 1]$ and strictly decreasing on $[1, \infty)$.

Proof. Let $A = \{0, 3\}$, $B = \{1, 2\}$ and $C = \{4, 5\}$ be the vertex orbits. The vertices in A have degree three and the others degree two. Write a, b, c, d, f for the common masses of directed edges of types AB, AC, BA, BB, CA, respectively. Their multiplicities are $2, 4, 2, 2, 4$. Put

$$E = 361\mu^2 + 10488\mu + 59951 > 0.$$

The following numerators over the common denominator $36E$ give the stationary masses:

Mass	Numerator
a	$893\mu^2 + 26754\mu + 151753$
b	$893\mu^2 + 25944\mu + 152563$
c	$12(76\mu^2 + 2283\mu + 13091)$
d	$3(361\mu^2 + 9918\mu + 51521)$
f	$12(76\mu^2 + 2208\mu + 12941)$

Indeed, these entries are positive, $2a + 4b + 2c + 2d + 4f = 1$, and direct substitution verifies (1), which now reduces to

$$a = \frac{1}{180} + \frac{9(\mu c + 2f)}{10(\mu + 2)}, \qquad b = \frac{1}{180} + \frac{9(c + (\mu + 1)f)}{10(\mu + 2)},$$
$$c = \frac{1}{120} + \frac{9(\mu a + d)}{10(\mu + 1)}, \qquad d = \frac{1}{120} + \frac{9(a + \mu d)}{10(\mu + 1)},$$
$$f = \frac{1}{120} + \frac{9b}{10}.$$

The vertex scores are $\pi_A = a + 2b$, $\pi_B = c + d$, $\pi_C = 2f$:

$$\pi_A = \frac{893\mu^2 + 26214\mu + 152293}{12E},$$
$$\pi_B = \frac{5(133\mu^2 + 3810\mu + 20777)}{12E},$$
$$\pi_C = \frac{2(76\mu^2 + 2208\mu + 12941)}{3E}.$$

For all $\mu \geq 0$, their order is strict, since

$$\pi_A - \pi_B = \frac{19\mu^2 + 597\mu + 4034}{E} > 0, \qquad \pi_B - \pi_C = \frac{19\mu^2 + 462\mu + 119}{4E} > 0.$$

Thus the spread $s(\mu) = \max_i \pi_\mu(i) - \min_i \pi_\mu(i)$ is

$$s(\mu) = \frac{5(19\mu^2 + 570\mu + 3251)}{4E}, \qquad s'(\mu) = -\frac{855(\mu - 1)(19\mu + 221)}{2E^2}. \tag{6}$$

This derivative proves both asserted strict inequalities. For finite rational certificates one may also use

$$s(1) - s(1/2) = \frac{57}{20543092} > 0, \qquad s(1) - s(2) = \frac{57}{6479852} > 0.$$

□

The source's displayed spread inequality has the increasing direction, while its original motivating prose describes a shrinking range. Theorem 2 refutes both directions. Theorem 1 independently refutes coordinate monotonicity. The result therefore does not depend on a sign or transcription discrepancy.

Tasks 2–5. Proposed follow-ups

Task 2. Characterize connected graphs for which every coordinate is monotonic in μ for uniform personalization and every damping parameter in $(0, 1)$.

Task 3. For the house and theta graphs, determine exactly how the number and location of turning points vary with damping and with the personalization vector.

Task 4. Bound the number of positive critical points of one μ-PageRank coordinate in terms of graph size or the number of orbits of directed edges.

Task 5. Find conditions under which ∞-PageRank still correctly predicts the ordering between standard and non-backtracking scores, without assuming monotonicity along the whole parameter interval.

Verification and audit status. Four exact verification groups check the symbolic stationary certificates, the derivative and ordering identities, independent rational Gaussian elimination, and finite Neumann-series enclosures. The complete current 22-page preprint was read. The identified journal final could not be retrieved in full, so campaign eligibility remains pending and this package receives zero new credit. Review is by the authoring agent only.

References

[1] C. Glover, T. Jones, M. Kempton and A. Oveson, *Effects of Backtracking on PageRank*, arXiv:2211.13353v2, February 9, 2026. Definition 2.4 and Conjecture 3.1. Published article: Applied Mathematics and Computation **531** (2026), 130182, doi:10.1016/j.amc.2026.130182 (identified final; complete text not accessed).